UNIT 5 - COMPLETE NOTES

Factorization

Multiplication in reverse: taking out common factors, grouping, the square and cube formulas, and splitting the middle term of a trinomial.

What this unit covers

- **5.1** Factorization by taking out the common factor
- **5.2** Factorization by grouping
- **5.3** Difference of two squares and perfect-square trinomials
- **5.4** Sum and difference of two cubes
- **5.5** Trinomials of the form $x^2 + bx + c$
- **5.6** Trinomials of the form $ax^2 + bx + c$

How to use these notes

Learn the methods in order - each one is used inside the next. Work through every worked example with pen and paper, then check your answer by multiplying the factors back together. Yellow boxes hold the rules most often tested in board papers. The last two pages give solved mixed examples, board-style multiple-choice questions and short questions for self-testing.

Exam focus: Board papers regularly ask for (1) factorization by common factor and grouping, (2) using $a^2 - b^2$, (3) perfect-square trinomials, (4) sum and difference of cubes, and (5) trinomials solved by splitting the middle term.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 5: Factorization, starting at book page 126. Notes prepared by Shah Fahad.

1. What factorization is - and the common factor

Multiplying expands brackets: 3(x + 2) = 3x + 6. **Factorization** does the reverse - it writes an expression as a **product of its factors**: 3x + 6 = 3(x + 2). Just as 12 = 3 x 4 breaks a number into factors, factorization breaks a polynomial into simpler polynomials whose product is the original.

> **Factorization:** Expressing a polynomial as a product of two or more polynomials is called factorization. The factorization is **complete** when no factor can be broken down further.

Multiplication: $(x + 3)(x + 2) = x^2 + 5x + 6$ | Factorization: $x^2 + 5x + 6 = (x + 3)(x + 2)$

Method 1 - Take out the highest common factor (HCF)

If every term shares a common factor, take out the **highest** one first. For numbers, take the HCF; for variables, take the lowest power present in every term. This is always the first step, whatever method follows.

Example 1. Factorize $6x^2 + 9x$.
Solution: Both terms are divisible by 3x. $6x^2 + 9x$ = **3x(2x + 3)**. Check: 3x x 2x = $6x^2$ and 3x x 3 = 9x.

Example 2. Factorize $12a^2b - 18ab^2 + 6ab$.
Solution: The HCF of 12, 18 and 6 is 6; every term contains a and b to power 1. Taking out 6ab: = **6ab(2a - 3b + 1)**. Note the 1 left by the last term - never leave an empty place.

Example 3. Factorize 5x(x - 4) + 3(x - 4).
Solution: The whole bracket (x - 4) is a common factor: = **(x - 4)(5x + 3)**.

> **Always check first:** Before trying any other method, ask - *does every term share a factor?* Missing this step is the most common reason a factorization is marked incomplete in board papers.

2. Factorization by grouping

Some expressions have no factor common to *all* terms, but the terms can be arranged in **pairs**, and each pair has its own common factor. If the two pairs then produce the **same bracket**, that bracket is itself a common factor.

> **The grouping pattern:** ka + kb + ma + mb = k(a + b) + m(a + b) = **(a + b)(k + m)**. Grouping works whenever the first pair and the second pair share a binomial factor.

Steps

1. Split the four terms into two pairs.
2. Take the common factor out of each pair separately.
3. Take out the common bracket as a factor.
4. If the brackets do not match, rearrange the terms and try again.

Example 4. Factorize ax + ay + bx + by.
Solution: Group: a(x + y) + b(x + y) = **(x + y)(a + b)**.

Example 5. Factorize x^2 + 3x + 2x + 6.
Solution: Group: x(x + 3) + 2(x + 3) = **(x + 3)(x + 2)**. This is exactly the trinomial x^2 + 5x + 6 - grouping is the engine behind splitting the middle term in Sections 5 and 6.

Example 6. Factorize 2xy - 4y + 3x - 6.
Solution: Group: 2y(x - 2) + 3(x - 2) = **(x - 2)(2y + 3)**. Watch the signs: -4y and -6 both give the bracket (x - 2), not (x + 2).

Example 7. Factorize a^2 - ab + ac - bc.
Solution: Group: a(a - b) + c(a - b) = **(a - b)(a + c)**.

> **Sign trap:** When the second pair starts with a minus sign, take out a negative factor so the brackets match. Example: x^2 - 2x + 3x - 6 is fine as written, but x^2 + 2x - 3x - 6 needs x(x + 2) - 3(x + 2) - the minus is taken out with the 3.

3. Difference of squares and perfect squares

The algebraic formulas of Unit 4, read backwards, become factorization formulas. Two shapes must be recognised on sight.

Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

Clues: exactly **two terms**, joined by a **minus**, and **both terms are perfect squares**. Numbers that are squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

Example 8. Factorize (a) $x^2 - 49$ (b) $4a^2 - 25b^2$ (c) $16 - y^2$

Solution: (a) $x^2 - 7^2$ = **(x + 7)(x - 7)**. (b) $(2a)^2 - (5b)^2$ = **(2a + 5b)(2a - 5b)**. (c) $4^2 - y^2$ = **(4 + y)(4 - y)**.

Example 9. Factorize $3x^2 - 27$.

Solution: Common factor first: $3(x^2 - 9)$ = **3(x + 3)(x - 3)**.

Perfect-square trinomials

$$a^2 + 2ab + b^2 = (a + b)^2 \quad \text{and} \quad a^2 - 2ab + b^2 = (a - b)^2$$

Clues: **three terms**; the first and last are perfect squares; the middle term is **twice the product** of their square roots (2ab). The sign of the middle term decides the sign inside the bracket.

Example 10. Factorize $x^2 + 10x + 25$.

Solution: Square roots: x and 5. Check the middle: 2(x)(5) = 10x. Correct, so = $\mathbf{(x + 5)^2}$.

Example 11. Factorize $9a^2 - 12ab + 4b^2$.

Solution: Square roots: 3a and 2b. Check: 2(3a)(2b) = 12ab, with a minus sign. So = $\mathbf{(3a - 2b)^2}$.

Not factorable this way: $a^2 + b^2$ (a *sum* of squares) does not factorize over real numbers, and $x^2 + 8x + 25$ is not a perfect square because 2(x)(5) = 10x, not 8x. Always run the middle-term check before writing the squared bracket.

4. Sum and difference of two cubes

When both terms are perfect **cubes** - like 1, 8, 27, 64, 125, or x^3, $8a^3$, $27b^3$ - two more formulas apply, for both the sum and the difference.

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

How to remember the signs

- The **first bracket** keeps the sign of the original expression: plus stays plus, minus stays minus.
- In the second bracket, the **middle term takes the opposite sign**, and the **last term is always positive**.
- The second bracket is built as: square of the first, product of the two, square of the second.

Example 12. Factorize $x^3 + 27$.
Solution: $27 = 3^3$, so a = x and b = 3: $= (x + 3)(x^2 - 3x + 9) =$ **$(x + 3)(x^2 - 3x + 9)$**.

Example 13. Factorize $8a^3 - 125$.
Solution: $8a^3 = (2a)^3$ and $125 = 5^3$, so a = 2a and b = 5 in the formula: $= (2a - 5)((2a)^2 + (2a)(5) + 5^2)$ $=$ **$(2a - 5)(4a^2 + 10a + 25)$**.

Example 14. Factorize $27x^3 + 8y^3$.
Solution: Cube roots are 3x and 2y: $= (3x + 2y)((3x)^2 - (3x)(2y) + (2y)^2) =$ **$(3x + 2y)(9x^2 - 6xy + 4y^2)$**.

Example 15. Factorize $2x^3 - 16$.
Solution: Common factor first: $2(x^3 - 8) = 2(x - 2)(x^2 + 2x + 4) =$ **$2(x - 2)(x^2 + 2x + 4)$**.

> **Stop there:** The quadratic factor $a^2 - ab + b^2$ or $a^2 + ab + b^2$ does **not** factorize further over real numbers. Do not try to split its middle term - the factorization is already complete.

5. Trinomials of the form $x^2 + bx + c$

A **trinomial** has three terms. When the coefficient of x^2 is 1, the factorization has a simple shape:

$$x^2 + bx + c = (x + p)(x + q), \quad \text{where} \quad p + q = b \quad \text{and} \quad pq = c$$

> **The rule:** Find two numbers whose **product is c** (the constant term) and whose **sum is b** (the coefficient of x). List the factor pairs of c first, then test which pair adds up to b.

Sign guide

Sign of c	Sign of b	Signs of p and q
c positive	b positive	Both positive: (+)(+)
c positive	b negative	Both negative: (-)(-)
c negative	either	Opposite signs; the larger number takes the sign of b

Example 16. Factorize $x^2 + 7x + 12$.
Solution: Factor pairs of 12: (1, 12), (2, 6), (3, 4). The pair 3 and 4 adds to 7. So = **(x + 3)(x + 4)**.

Example 17. Factorize $x^2 - 8x + 15$.
Solution: Product +15, sum -8: both numbers negative, -3 and -5. So = **(x - 3)(x - 5)**.

Example 18. Factorize $x^2 + 2x - 24$.
Solution: Product -24, sum +2: opposite signs, and the larger is positive: +6 and -4, since 6 + (-4) = 2 and 6 x (-4) = -24. So = **(x + 6)(x - 4)**.

Example 19. Factorize $x^2 - 5x - 14$.
Solution: Product -14, sum -5: -7 and +2. So = **(x - 7)(x + 2)**.

> **If no pair works:** If no factor pair of c adds up to b, the trinomial does not factorize with whole numbers. Say so - do not force an answer. Example: $x^2 + 3x + 1$ has no such pair.

6. Trinomials of the form $ax^2 + bx + c$

When the coefficient of x^2 is not 1, the two numbers must be found from **a x c**, not from c alone. The method is called **splitting the middle term**.

> **The rule:** Find two numbers m and n such that **m + n = b** and **m x n = a x c**. Rewrite bx as mx + nx, then factorize the four terms by grouping (Section 2).

Steps

1. Multiply a x c.
2. Find two numbers with product ac and sum b.
3. Split the middle term bx into those two parts.
4. Group in pairs and take out the common factors.

Example 20. Factorize $2x^2 + 7x + 3$.

Solution: ac = 2 x 3 = 6. Two numbers with product 6 and sum 7: 6 and 1. Split: $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) =$ **$(x + 3)(2x + 1)$**.

Example 21. Factorize $3x^2 - 10x + 8$.

Solution: ac = 3 x 8 = 24. Product 24, sum -10: -6 and -4. Split: $3x^2 - 6x - 4x + 8 = 3x(x - 2) - 4(x - 2) =$ **$(x - 2)(3x - 4)$**.

Example 22. Factorize $6x^2 + x - 12$.

Solution: ac = 6 x (-12) = -72. Product -72, sum +1: +9 and -8. Split: $6x^2 + 9x - 8x - 12 = 3x(2x + 3) - 4(2x + 3) =$ **$(2x + 3)(3x - 4)$**.

Example 23. Factorize $4x^2 - 12x + 8$.

Solution: Common factor first: $4(x^2 - 3x + 2)$. Now the inside is the easy type: numbers -1 and -2 give product +2 and sum -3. So = **$4(x - 1)(x - 2)$**. Taking the common factor out first turned a hard trinomial into an easy one.

> **Check by multiplying back:** $(2x + 3)(3x - 4) = 6x^2 - 8x + 9x - 12 = 6x^2 + x - 12$. Correct. A 20-second check like this catches almost every sign and split error.

7. Special forms and the complete strategy

A form worth recognising: $a^4 + a^2b^2 + b^4$

This expression looks like a perfect square but the middle term is too small. Complete the square by adding and subtracting a^2b^2, then use the difference of squares:

$$a^4 + a^2b^2 + b^4 = (a^2 + b^2)^2 - (ab)^2 = (a^2 + ab + b^2)(a^2 - ab + b^2)$$

Example 24. Factorize $x^4 + x^2y^2 + y^4$.
Solution: Add and subtract x^2y^2: $(x^4 + 2x^2y^2 + y^4) - x^2y^2 = (x^2 + y^2)^2 - (xy)^2 =$ **$(x^2 + xy + y^2)(x^2 - xy + y^2)$**.

The complete strategy - use this order every time

Step	Ask	Method
1	Is there a common factor?	Take out the HCF - always first.
2	How many terms are left?	2 terms: difference of squares, or sum/difference of cubes. 3 terms: perfect square, or split the middle term. 4 terms: grouping.
3	Can any factor be broken further?	Repeat the checks on each factor until none can be split.
4	Does it check?	Multiply the factors back and compare with the question.

Example 25. Factorize completely: $2x^3 - 18x$.
Solution: Step 1 - common factor 2x: $2x(x^2 - 9)$. Step 2 - difference of squares inside: = **$2x(x + 3)(x - 3)$**. Nothing can be split further, so this is complete.

Example 26. Factorize completely: $x^4 - 81$.
Solution: Difference of squares twice: $(x^2 + 9)(x^2 - 9) =$ **$(x^2 + 9)(x + 3)(x - 3)$**. The factor $x^2 + 9$ is a sum of squares and stays as it is.

Complete means complete: An answer like $2x(x^2 - 9)$ loses marks, because $x^2 - 9$ can still be factorized. Keep going until every factor is in its simplest form.

8. Solved examples - mixed practice

Example 27. Factorize $15a^2b - 10ab^2$.
Solution: HCF is 5ab: = **5ab(3a - 2b)**.

Example 28. Factorize px + py - qx - qy.
Solution: Group: p(x + y) - q(x + y) = **(x + y)(p - q)**.

Example 29. Factorize $25m^2 - 36n^2$.
Solution: Difference of squares, $(5m)^2 - (6n)^2$: = **(5m + 6n)(5m - 6n)**.

Example 30. Factorize $a^2 + 14a + 49$.
Solution: Check: 2(a)(7) = 14a, so it is a perfect square: = **$(a + 7)^2$**.

Example 31. Factorize $y^3 - 64$.
Solution: Difference of cubes, $64 = 4^3$: = **$(y - 4)(y^2 + 4y + 16)$**.

Example 32. Factorize $x^2 - x - 20$.
Solution: Product -20, sum -1: -5 and +4. So = **(x - 5)(x + 4)**.

Example 33. Factorize $5x^2 + 16x + 3$.
Solution: ac = 15; product 15, sum 16: 15 and 1. Split: $5x^2 + 15x + x + 3 = 5x(x + 3) + 1(x + 3)$ = **(x + 3)(5x + 1)**.

Common mistakes to avoid

- Stopping early - leaving a factor like $(x^2 - 9)$ unsplit. Factorize **completely**.
- Trying to factorize $a^2 + b^2$. A sum of squares has no real factorization.
- Writing $x^2 + 5x + 6 = (x + 5)(x + 6)$. The numbers must give the correct **sum and product** - here 2 and 3, not 5 and 6.
- Wrong signs in the cube formulas: the middle term of the second bracket takes the **opposite** sign.
- Forgetting the 1 when a whole term is taken out as a common factor: 6ab(2a - 3b + 1), not 6ab(2a - 3b).

9. Test yourself - MCQs and short questions

Multiple-choice questions

1. The HCF factorization of $8x^2 + 12x$ is: (a) $2x(4x + 6)$ (b) $4x(2x + 3)$ (c) $4(2x^2 + 3x)$ (d) $x(8x + 12)$
   Answer: (b) - highest common factor 4x.
2. $x^2 - 16$ equals: (a) $(x - 4)^2$ (b) $(x + 4)^2$ (c) $(x + 4)(x - 4)$ (d) $(x + 16)(x - 1)$
   Answer: (c) - difference of squares.
3. $x^2 + 6x + 9$ equals: (a) $(x + 3)^2$ (b) $(x + 9)(x + 1)$ (c) $(x - 3)^2$ (d) $(x + 6)(x + 9)$
   Answer: (a) - $2(x)(3) = 6x$.
4. $x^3 - 8$ equals: (a) $(x - 2)(x^2 + 2x + 4)$ (b) $(x - 2)^3$ (c) $(x + 2)(x^2 - 2x + 4)$ (d) $(x - 8)(x^2 + x + 1)$
   Answer: (a) - difference of cubes.
5. $x^2 + 9x + 20$ equals: (a) $(x + 2)(x + 10)$ (b) $(x + 4)(x + 5)$ (c) $(x + 20)(x + 1)$ (d) $(x - 4)(x - 5)$
   Answer: (b) - $4 + 5 = 9$ and $4 \times 5 = 20$.
6. $x^2 - 3x - 10$ equals: (a) $(x - 2)(x + 5)$ (b) $(x + 2)(x + 5)$ (c) $(x - 5)(x + 2)$ (d) $(x - 10)(x + 1)$
   Answer: (c) - $-5 + 2 = -3$ and $-5 \times 2 = -10$.
7. $2x^2 + 5x + 2$ equals: (a) $(2x + 1)(x + 2)$ (b) $(2x + 2)(x + 1)$ (c) $(x + 5)(2x + 2)$ (d) $(2x - 1)(x - 2)$
   Answer: (a) - $ac = 4$, split $5x$ as $4x + x$.
8. Which expression does NOT factorize over real numbers? (a) $x^2 - 9$ (b) $x^2 + 9$ (c) $x^2 - 6x + 9$ (d) $x^3 + 27$
   Answer: (b) - a sum of squares.

Short questions for written practice

1. Factorize by taking out the common factor: (a) $9x^2y + 6xy^2$ (b) $4a - 8b + 12c$. (Answers: $3xy(3x + 2y)$; $4(a - 2b + 3c)$)
2. Factorize by grouping: $ab + ac + b^2 + bc$. (Answer: $(b + c)(a + b)$)
3. Factorize: (a) $49 - x^2$ (b) $x^2 - 14x + 49$. (Answers: $(7 + x)(7 - x)$; $(x - 7)^2$)
4. Factorize: (a) $x^3 + 1$ (b) $64a^3 - b^3$. (Answers: $(x + 1)(x^2 - x + 1)$; $(4a - b)(16a^2 + 4ab + b^2)$)
5. Factorize: (a) $x^2 + 11x + 30$ (b) $x^2 - 2x - 35$. (Answers: $(x + 5)(x + 6)$; $(x - 7)(x + 5)$)
6. Factorize completely: (a) $3x^2 - 11x + 6$ (b) $3x^3 - 3x$. (Answers: $(x - 3)(3x - 2)$; $3x(x + 1)(x - 1)$)

> **One-page revision:** Always take out the HCF first. Two terms: $a^2 - b^2 = (a + b)(a - b)$; $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$; $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. Three terms: $a^2 +/- 2ab + b^2 = (a +/- b)^2$; otherwise find two numbers - product c, sum b for $x^2 + bx + c$; product ac, sum b for $ax^2 + bx + c$ - and split the middle term. Four terms: group in pairs. Finish by multiplying back to check.

End of Unit 5 notes. Next unit: Unit 6 - Algebraic Manipulation. Source book: A Textbook of Mathematics for Grade IX, KPTB Peshawar.